Online Appendix for ‘Macroeconomic Stabilization and Endogenous Inequality in a TANK Model with Dual Nominal Rigidities’

This online appendix reports the nonlinear primitives of the model and the intermediate log-linearization steps underlying Section 3 of the main text. Appendix A in the main text derives the analytical reduction from the nonredundant 20-equation system to the five-equation canonical representation.

Model Description

The appendix first states the nonlinear problems of the labor packer, households, firms, and government. It then collects a reduced nonlinear equilibrium system sufficient for the first-order approximation, and finally reports the nonredundant log-linear system used in Section 3 of the main text.

Throughout, upper-case letters denote level variables. The index $h \in [0,1]$ refers to households, and the index $j \in [0,1]$ refers to intermediate-goods firms. A superscript $i \in \{H, S\}$ indicates the household type, where (H) denotes hand-to-mouth households and S denotes saver households. P_t denotes the final-good price index, W_t^N the aggregate nominal wage, and $W_t \equiv W_t^N/P_t$ the aggregate real wage. For each household type i, C_t^i, N_t^i, and $W_t^{N,i}$ denote consumption, labor supply, and the nominal wage, while $W_t^i \equiv W_t^{N,i}/P_t$ denotes the corresponding real wage. Aggregate price inflation is defined as $\Pi_t^p \equiv P_t/P_{t-1}$, and type-specific wage inflation is defined as $\Pi_t^i \equiv W_t^{N,i}/W_{t-1}^{N,i}$. Preferences are represented by a general period utility function $U(C,N)$, with marginal utilities $U_C(C,N) > 0$ and $U_N(C,N) < 0$. The marginal rate of substitution is $MRS(C,N) \equiv -U_N(C,N)/U_C(C,N)$. Intermediate firms use the general technology $Y = A_t F(N)$, with $F_N(N) > 0$. The type-i wage markup is $MU_t^i \equiv W_t^i/MRS(C_t^i, N_t^i)$, and the price markup is $MU_t^p \equiv A_t F_N(N_t)/W_t$. Finally, R_t^N denotes the gross nominal interest rate and D_t firm dividends. Here T_t^i denotes the net fiscal liability of type-i households. A negative value therefore corresponds to a transfer received by that household type. The variable Z_t denotes the transfer directed to Type H households, and z_t is the redistribution shock that drives Z_t through $Z_t = z_t Y$.

Labor Packer

The labor packer is a competitive aggregator that combines differentiated household labor into a homogeneous labor input sold to intermediate-goods firms. Household h supplies labor $N_t(h)$ and is paid the nominal wage $W_t^N(h)$, while the packer sells the aggregate labor input N_t at the aggregate nominal wage W_t^N. The parameter $\psi_w > 1$ is the elasticity of substitution across labor varieties, and Λ_t^L is the multiplier on the constant-elasticity-of-substitution aggregation constraint. Taking the household-specific wages $\{W_t^N(h)\}_{h\in[0,1]}$ as given, the labor packer chooses $\{N_t(h)\}_{h\in[0,1]}$ and N_t, which yields the labor demand schedule for each household and the aggregate nominal wage index.

- Objects taken as given: $P_t, W_t^N, \{W_t^N(h)\}_{h\in[0,1]}$
- Choice variables: $N_t, \{N_t(h)\}_{h\in[0,1]}$
- Objective Function:

$$W_t^N N_t - \int_0^1 W_t^N(h) N_t(h) dh$$

- Constraints:

$$N_t = \left(\int_0^1 N_t(h)^{\frac{\psi_w-1}{\psi_w}} dh\right)^{\frac{\psi_w}{\psi_w-1}}, \ \psi_w > 1.$$

- Lagrangian:

$$\begin{aligned}\mathcal{L}_t =& W_t^N N_t - \int_0^1 W_t^N(h) N_t(h) dh \\ &+ \Lambda_t^L \left(\left(\int_0^1 N_t(h)^{\frac{\psi_w-1}{\psi_w}} dh\right)^{\frac{\psi_w}{\psi_w-1}} - N_t\right)\end{aligned}$$

- FOC wrt N_t:

$$W_t^N = \Lambda_t^L$$

- FOC wrt $N_t(h)$:

$$N_t(h) = N_t \left(\frac{W_t^N(h)}{W_t^N}\right)^{-\psi_w}$$

- Nominal wage:

$$W_t^N = \left[\int_0^1 W_t^N(h)^{1-\psi_w} dh\right]^{\frac{1}{1-\psi_w}}$$

Households

There is a continuum of households indexed by $h \in [0,1]$. Each household consumes $C_t(h)$, supplies differentiated labor $N_t(h)$, and sets its own nominal wage $W_t^N(h)$. Preferences are common across households and are represented by period utility $U(C_t(h), N_t(h))$. Wage setting is subject to Rotemberg-type adjustment costs governed by η_w, and labor income is subsidized at rate τ_w. A fraction λ of households are hand-to-mouth households (Type H), who do not trade assets, while the remaining fraction $1-\lambda$ are saver households (Type S), who hold one-period nominal bonds and receive firm dividends. Within each type, full risk sharing implies identical consumption, labor, and wage choices in equilibrium.

With two household types and within-type symmetry, the labor-packer aggregation functions defined and derived in the previous subsection can be rewritten as follows:

$$\begin{aligned} N_t &= \left[\lambda (N_t^H)^{\frac{\psi_w-1}{\psi_w}} + (1-\lambda)(N_t^S)^{\frac{\psi_w-1}{\psi_w}}\right]^{\frac{\psi_w}{\psi_w-1}} \\ W_t^N &= \left[\lambda (W_t^{N,H})^{1-\psi_w} + (1-\lambda)(W_t^{N,S})^{1-\psi_w}\right]^{\frac{1}{1-\psi_w}} \\ &= P_t \left[\lambda (W_t^H)^{1-\psi_w} + (1-\lambda)(W_t^S)^{1-\psi_w}\right]^{\frac{1}{1-\psi_w}} \end{aligned}$$

Type H

A Type H household chooses current consumption $C_t(h)$, labor supply $N_t(h)$, and its own nominal wage $W_t^N(h)$, taking aggregate variables as given. Since it does not trade financial assets, its current uses of funds consist of consumption and the net fiscal liability T_t^H, where $T_t^H < 0$ corresponds to a transfer received from the government. In the budget constraint, P_t is the aggregate price level, Y_t is aggregate output, and η_w measures the cost of changing the nominal wage relative to the previous period's wage $W_{t-1}^N(h)$. The optimality conditions imply the marginal utility of consumption, the household-specific labor demand schedule, and the Type H wage-setting equation. In symmetric equilibrium, these conditions can be written in terms of the Type H real wage W_t^H, wage inflation Π_t^H, marginal rate of substitution $MRS_t^H \equiv MRS(C_t^H, N_t^H)$, and wage markup $MU_t^H \equiv W_t^H / MRS_t^H$.

- Objects taken as given: $W_t^N, P_t, N_t, Y_t, T_t^H, W_{t-1}^N(h)$
- Choice variables: $C_t(h), N_t(h), W_t^N(h)$
- Objective Function:

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t U(C_t(h), N_t(h))$$

- Constraints:

$$
\begin{aligned}
& C_t(h) + T_t^H \\
& = (1+\tau_w)\frac{W_t^N(h)}{P_t} N_t(h) - \frac{\eta_w}{2}\left(\frac{W_t^N(h)}{W_{t-1}^N(h)} - 1\right)^2 Y_t \\
& N_t(h) = N_t\left(\frac{W_t^N(h)}{W_t^N}\right)^{-\psi_w}
\end{aligned}
$$

- Lagrangian:

$$
\begin{aligned}
\mathbb{L} =& \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \mathcal{L}_t \\
\mathcal{L}_t =& U\left(C_t(h), N_t\left(\frac{W_t^N(h)}{W_t^N}\right)^{-\psi_w}\right) \\
& + \Lambda_t^H(h)\left(\begin{array}{c} (1+\tau_w)\frac{N_t}{P_t} W_t^N \left(\frac{W_t^N(h)}{W_t^N}\right)^{1-\psi_w} - \frac{\eta_w}{2}\left(\frac{W_t^N(h)}{W_{t-1}^N(h)} - 1\right)^2 Y_t \\ -C_t(h) - T_t^H \end{array}\right)
\end{aligned}
$$

- FOC wrt $C_t(h)$:

$$
\Lambda_t^H(h) = U_C(C_t(h), N_t(h))
$$

- FOC wrt $W_t^N(h)$:

$$
\begin{aligned}
-\psi_w \frac{N_t}{W_t^N}\left(\frac{W_t^N(h)}{W_t^N}\right)^{-\psi_w - 1} U_N(C_t(h), N_t(h)) \\
+(1-\psi_w)\Lambda_t^H(h)(1+\tau_w)\frac{N_t}{P_t}\left(\frac{W_t^N(h)}{W_t^N}\right)^{-\psi_w} \\
-\Lambda_t^H(h)\eta_w\left(\frac{W_t^N(h)}{W_{t-1}^N(h)} - 1\right)\frac{Y_t}{W_{t-1}^N(h)} \\
+\beta\mathbb{E}_t\Lambda_{t+1}^H(h)\eta_w\left(\frac{W_{t+1}^N(h)}{W_t^N(h)} - 1\right)\frac{W_{t+1}^N(h) Y_{t+1}}{W_t^N(h)^2} = 0
\end{aligned}
$$

- Symmetric Equilibrium ($\Lambda_t^H(h) = \Lambda_t^H$, $C_t(h) = C_t^H$, $N_t(h) = N_t^H$, $W_t^N(h) = W_t^{N,H}$):

$$
\begin{aligned}
N_t^H &= N_t\left(\frac{W_t^{N,H}}{W_t^N}\right)^{-\psi_w} \\
\Lambda_t^H &= U_C(C_t^H, N_t^H)
\end{aligned}
$$

$$
\begin{aligned}
\eta_w \frac{Y_t}{N_t^H}\frac{W_t^{N,H}}{W_{t-1}^{N,H}}\left(\frac{W_t^{N,H}}{W_{t-1}^{N,H}}-1\right) =& \psi_w\left[MRS_t^H-(1+\tau_w)\left(1-\frac{1}{\psi_w}\right)\frac{W_t^{N,H}}{P_t}\right] \\
&+\beta\eta_w\mathbb{E}_t\left[\frac{U_C(C_{t+1}^H,N_{t+1}^H)}{U_C(C_t^H,N_t^H)}\frac{Y_{t+1}}{N_t^H}\frac{W_{t+1}^{N,H}}{W_t^{N,H}}\left(\frac{W_{t+1}^{N,H}}{W_t^{N,H}}-1\right)\right] \\
\Leftrightarrow \eta_w\frac{W_t^{N,H}}{W_{t-1}^{N,H}}\left(\frac{W_t^{N,H}}{W_{t-1}^{N,H}}-1\right) =& \psi_w\frac{W_t^{N,H}N_t^H}{P_tY_t}\left[\frac{1}{MU_t^H}-(1+\tau_w)\left(1-\frac{1}{\psi_w}\right)\right] \\
&+\beta\eta_w\mathbb{E}_t\left[\frac{U_C(C_{t+1}^H,N_{t+1}^H)}{U_C(C_t^H,N_t^H)}\frac{Y_{t+1}}{Y_t}\frac{W_{t+1}^{N,H}}{W_t^{N,H}}\left(\frac{W_{t+1}^{N,H}}{W_t^{N,H}}-1\right)\right]
\end{aligned}
$$

Type S

A Type S household chooses consumption $C_t(h)$, labor supply $N_t(h)$, its own nominal wage $W_t^N(h)$, and one-period nominal bond holdings $B_t^N(h)$. In addition to current after-subsidy labor income, it receives gross nominal bond returns $R_{t-1}^N B_{t-1}^N(h)$, firm dividends D_t, and faces the net fiscal liability T_t^S. A positive value of T_t^S is therefore a levy, while a negative value corresponds to a transfer. Here R_t^N denotes the gross nominal interest rate on the bond purchased in period t, and $B_t^N(h)/P_t$ is the household's real bond position. The optimality conditions deliver both an Euler equation governing intertemporal consumption smoothing and a wage-setting equation analogous to that of Type H households. In symmetric equilibrium, these conditions are summarized by the Type S real wage W_t^S, wage inflation Π_t^S, marginal rate of substitution $MRS_t^S \equiv MRS(C_t^S, N_t^S)$, and wage markup $MU_t^S \equiv W_t^S/MRS_t^S$.

- Objects taken as given: $W_t^N, P_t, R_t^N, D_t, N_t, Y_t, T_t^S, W_{t-1}^N(h), B_{t-1}^N(h)$
- Choice variables: $C_t(h), N_t(h), W_t^N(h), B_t^N(h)$
- Objective Function:
$$\mathbb{E}_0\sum_{t=0}^{\infty}\beta^t U(C_t(h),N_t(h))$$
- Constraints:
$$
\begin{aligned}
&C_t(h)+\frac{B_t^N(h)}{P_t}+T_t^S \\
&=(1+\tau_w)\frac{W_t^N(h)}{P_t}N_t(h)-\frac{\eta_w}{2}\left(\frac{W_t^N(h)}{W_{t-1}^N(h)}-1\right)^2 Y_t \\
&+R_{t-1}^N\frac{B_{t-1}^N(h)}{P_t}+D_t \\
&N_t(h)=N_t\left(\frac{W_t^N(h)}{W_t^N}\right)^{-\psi_w}
\end{aligned}
$$

- Lagrangian:

$$
\begin{aligned}
\mathbb{L} =& \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \mathcal{L}_t \\
\mathcal{L}_t =& U\left(C_t(h), N_t \left(\frac{W_t^N(h)}{W_t^N} \right)^{-\psi_w} \right) \\
&+ \Lambda_t^S(h) \left(\begin{array}{c} (1+\tau_w) \frac{W_t^N N_t}{P_t} \left(\frac{W_t^N(h)}{W_t^N} \right)^{1-\psi_w} - \frac{\eta_w}{2} \left(\frac{W_t^N(h)}{W_{t-1}^N(h)} - 1 \right)^2 Y_t \\ + R_{t-1}^N \frac{B_{t-1}^N(h)}{P_t} + D_t - C_t(h) - \frac{B_t^N(h)}{P_t} - T_t^S \end{array} \right)
\end{aligned}
$$

- FOC wrt $C_t(h)$:

$$
\Lambda_t^S(h) = U_C(C_t(h), N_t(h))
$$

- FOC wrt $W_t^N(h)$:

$$
\begin{aligned}
-\psi_w \frac{N_t}{W_t^N} \left(\frac{W_t^N(h)}{W_t^N} \right)^{-\psi_w - 1} & U_N(C_t(h), N_t(h)) \\
+(1-\psi_w)\Lambda_t^S(h)(1+\tau_w) \frac{N_t}{P_t} & \left(\frac{W_t^N(h)}{W_t^N} \right)^{-\psi_w} \\
-\Lambda_t^S(h)\eta_w \left(\frac{W_t^N(h)}{W_{t-1}^N(h)} - 1 \right) & \frac{Y_t}{W_{t-1}^N(h)} \\
+\beta \mathbb{E}_t \Lambda_{t+1}^S(h) \eta_w \left(\frac{W_{t+1}^N(h)}{W_t^N(h)} - 1 \right) & \frac{W_{t+1}^N(h) Y_{t+1}}{W_t^N(h)^2} = 0
\end{aligned}
$$

- FOC wrt $B_t^N(h)$:

$$
\frac{\Lambda_t^S(h)}{P_t} = \beta \mathbb{E}_t \left[R_t^N \frac{\Lambda_{t+1}^S(h)}{P_{t+1}} \right]
$$

- Symmetric Equilibrium ($\Lambda_t^S(h) = \Lambda_t^S$, $C_t(h) = C_t^S$, $N_t(h) = N_t^S$, $W_t^N(h) = W_t^{N,S}$, where W_t^S is the real wage of type S):

$$
\begin{aligned}
N_t^S &= N_t \left(\frac{W_t^{N,S}}{W_t^N} \right)^{-\psi_w} \\
\Lambda_t^S &= U_C(C_t^S, N_t^S) \\
1 =& \beta \mathbb{E}_t \left[R_t^N \frac{U_C(C_{t+1}^S, N_{t+1}^S)}{U_C(C_t^S, N_t^S)} \frac{P_t}{P_{t+1}} \right]
\end{aligned}
$$

$$
\begin{aligned}
\eta_w \frac{Y_t}{N_t^S}\frac{W_t^{N,S}}{W_{t-1}^{N,S}}\left(\frac{W_t^{N,S}}{W_{t-1}^{N,S}}-1\right) =&\psi_w\left[MRS_t^S-(1+\tau_w)\left(1-\frac{1}{\psi_w}\right)\frac{W_t^{N,S}}{P_t}\right] \\
&+\beta\eta_w\mathbb{E}_t\left[\frac{U_C(C_{t+1}^S,N_{t+1}^S)}{U_C(C_t^S,N_t^S)}\frac{Y_{t+1}}{N_t^S}\frac{W_{t+1}^{N,S}}{W_t^{N,S}}\left(\frac{W_{t+1}^{N,S}}{W_t^{N,S}}-1\right)\right] \\
\Leftrightarrow \eta_w \frac{W_t^{N,S}}{W_{t-1}^{N,S}}\left(\frac{W_t^{N,S}}{W_{t-1}^{N,S}}-1\right) =&\psi_w\frac{W_t^{N,S}N_t^S}{P_tY_t}\left[\frac{1}{MU_t^S}-(1+\tau_w)\left(1-\frac{1}{\psi_w}\right)\right] \\
&+\beta\eta_w\mathbb{E}_t\left[\frac{U_C(C_{t+1}^S,N_{t+1}^S)}{U_C(C_t^S,N_t^S)}\frac{Y_{t+1}}{Y_t}\frac{W_{t+1}^{N,S}}{W_t^{N,S}}\left(\frac{W_{t+1}^{N,S}}{W_t^{N,S}}-1\right)\right]
\end{aligned}
$$

where $MRS_t^S = MRS(C_t^S, N_t^S)$ is Type S marginal rate of substitution, and $MU_t^S = W_t^S/MRS_t^S$ is Type S wage markup.

Final Goods Firm

The final-goods firm is perfectly competitive and combines differentiated intermediate goods into a homogeneous final good using a constant-elasticity-of-substitution aggregator. The quantity of variety j is denoted by $Y_t(j)$, its price by $P_t(j)$, aggregate final output by Y_t, and the final-good price index by P_t. The elasticity of substitution across intermediate varieties is $\psi_p \geq 1$, and Λ_t^F is the multiplier on the aggregation constraint. Taking the prices $\{P_t(j)\}_{j\in[0,1]}$ as given, the final-goods firm chooses $(\{Y_t(j)\}_{j\in[0,1]})$ and Y_t, which yields the demand schedule for each intermediate-good producer and the aggregate final-good price index.

- Objects taken as given: $P_t, \{P_t(j)\}_{j\in[0,1]}$
- Choice variables: $Y_t, \{Y_t(j)\}_{j\in[0,1]}$
- Objective Function:

$$
P_tY_t - \int_0^1 P_t(j)Y_t(j)dj
$$

- Constraints:

$$
Y_t = \left(\int_0^1 Y_t(j)^{\frac{\psi_p-1}{\psi_p}}dj\right)^{\frac{\psi_p}{\psi_p-1}}, \ \psi_p \geq 1
$$

- Lagrangian:

$$
\begin{aligned}
\mathcal{L}_t =&P_tY_t - \int_0^1 P_t(j)Y_t(j)dj \\
&+\Lambda_t^F\left(\left(\int_0^1 Y_t(j)^{\frac{\psi_p-1}{\psi_p}}dj\right)^{\frac{\psi_p}{\psi_p-1}} - Y_t\right)
\end{aligned}
$$

- FOC wrt Y_t:
$$\Lambda_t^F = P_t$$

- FOC wrt $Y_t(j)$:
$$Y_t(j) = Y_t \left(\frac{P_t(j)}{P_t} \right)^{-\psi_p}$$

- Price of Final Goods:
$$P_t = \left[\int_0^1 P_t(j)^{1-\psi_p} dj \right]^{\frac{1}{1-\psi_p}}$$

Intermediate Goods Firms

Intermediate-good firm j produces output $Y_t(j)$ using labor $N_t(j)$ according to the general technology $Y_t(j) = A_t F(N_t(j))$, where $A_t \equiv \exp(a_t)$ denotes aggregate labor productivity and $F_N(N) > 0$. The firm hires homogeneous labor from the labor packer at the nominal wage W_t^N, sets its own price $P_t(j)$, and faces Rotemberg price adjustment costs governed by η_p. Its real profits are denoted by $D_t(j)$, the government levies the lump-sum tax T_t^M, and the sales subsidy rate is τ_p. The firm's objective is the date-0 present discounted value of profits, discounted by the saver household's stochastic discount factor. The optimality conditions determine labor demand, real marginal cost, and the price-setting equation. In symmetric equilibrium, the price markup is $MU_t^p \equiv P_t MPN_t / W_t^N = MPN_t / W_t$ and $MPN_t \equiv A_t F_N(N_t)$ is the marginal product of labor.

- Objects taken as given: $W_t^N, P_t, Y_t, A_t, T_t^M, C_t^S, P_{t-1}(j)$
- Choice variables: $N_t(j), P_t(j), Y_t(j)$
- Objective Function:
$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \frac{U_C(C_t^S, N_t^S)}{U_C(C_0^S, N_0^S)} D_t(j)$$

- Constraints:
$$\begin{aligned}
D_t(j) &= (1+\tau_p) \frac{P_t(j)}{P_t} Y_t(j) - \frac{\eta_p}{2} \left(\frac{P_t(j)}{P_{t-1}(j)} - 1 \right)^2 Y_t - \frac{W_t^N}{P_t} N_t(j) - T_t^M \\
Y_t(j) &= Y_t \left(\frac{P_t(j)}{P_t} \right)^{-\psi_p} \\
Y_t(j) &= A_t F(N_t(j))
\end{aligned}$$

- Lagrangian:

$$
\begin{aligned}
\mathbb{L} =& \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \frac{U_C(C_t^S, N_t^S)}{U_C(C_0^S, N_0^S)} \mathcal{L}_t \\
\mathcal{L}_t =& (1+\tau_p) Y_t \left(\frac{P_t(j)}{P_t} \right)^{1-\psi_p} - \frac{\eta_p}{2} \left(\frac{P_t(j)}{P_{t-1}(j)} - 1 \right)^2 Y_t - \frac{W_t^N}{P_t} N_t(j) \\
& - T_t^M + \Lambda_t^M(j) \left(A_t F(N_t(j)) - Y_t \left(\frac{P_t(j)}{P_t} \right)^{-\psi_p} \right)
\end{aligned}
$$

- FOC wrt $N_t(j)$:

$$\Lambda_t^M(j) A_t F_N(N_t(j)) = W_t^N / P_t$$

- FOC wrt $P_t(j)$:

$$
\begin{aligned}
& U_C(C_t^S, N_t^S)(1+\tau_p)(1-\psi_p) \frac{Y_t}{P_t} \left(\frac{P_t(j)}{P_t} \right)^{-\psi_p} - U_C(C_t^S, N_t^S) \eta_p \frac{Y_t}{P_{t-1}(j)} \left(\frac{P_t(j)}{P_{t-1}(j)} - 1 \right) \\
& + U_C(C_t^S, N_t^S) \psi_p \Lambda_t^M(j) \frac{Y_t}{P_t} \left(\frac{P_t(j)}{P_t} \right)^{-\psi_p - 1} + \beta \eta_p \mathbb{E}_t \left[U_C(C_{t+1}^S, N_{t+1}^S) Y_{t+1} \frac{P_{t+1}(j)}{P_t(j)^2} \left(\frac{P_{t+1}(j)}{P_t(j)} - 1 \right) \right] = 0
\end{aligned}
$$

- Symmetric Equilibrium ($N_t(j) = N_t$, $\Lambda_t^M(j) = \Lambda_t^M$, $P_t(j) = P_t$, $Y_t(j) = Y_t$):

$$
\begin{aligned}
\eta_p \frac{P_t}{P_{t-1}} \left(\frac{P_t}{P_{t-1}} - 1 \right) =& \psi_p \left[\frac{W_t^N / P_t}{MPN_t} - (1+\tau_p) \left(1 - \frac{1}{\psi_p} \right) \right] \\
& + \beta \eta_p \mathbb{E}_t \left[\frac{U_C(C_{t+1}^S, N_{t+1}^S)}{U_C(C_t^S, N_t^S)} \frac{Y_{t+1}}{Y_t} \frac{P_{t+1}}{P_t} \left(\frac{P_{t+1}}{P_t} - 1 \right) \right] \\
=& \psi_p \left[\frac{1}{MU_t^p} - (1+\tau_p) \left(1 - \frac{1}{\psi_p} \right) \right] \\
& + \beta \eta_p \mathbb{E}_t \left[\frac{U_C(C_{t+1}^S, N_{t+1}^S)}{U_C(C_t^S, N_t^S)} \frac{Y_{t+1}}{Y_t} \frac{P_{t+1}}{P_t} \left(\frac{P_{t+1}}{P_t} - 1 \right) \right]
\end{aligned}
$$

- Profits:

$$D_t = (1+\tau_p) Y_t - \frac{\eta_p}{2} \left(\frac{P_t}{P_{t-1}} - 1 \right)^2 Y_t - \frac{W_t^N}{P_t} N_t - T_t^M$$

  When $(1+\tau_p)(1 - 1/\psi_p) = 1$, steady state price markup is one.

Government

The government performs three functions in the model. First, it provides the wage subsidy τ_w and the sales subsidy τ_p, eliminating steady-state markups under the baseline specialization introduced below. Second, it redistributes resources across household types through the lump-sum components T_t^H and T_t^S. Here Z_t denotes the transfer received by Type H households, while Type S households finance that transfer through a balanced-budget levy. Because T_t^i denotes net fiscal liability, the transfer implies $T_t^H < 0$ for the recipient household type and a positive levy on Type S through T_t^S. Third, the monetary authority sets the gross nominal interest rate R_t^N according to a Taylor rule that responds to gross price inflation Π_t^p and to the monetary policy shock m_t. The redistribution shock z_t follows an autoregressive process and determines Z_t through $Z_t = z_t Y$, where Y is steady-state output.

Fiscal policy:

$$
\begin{aligned}
T_t^H &= -Z_t + \tau_w W_t^H N_t^H \\
T_t^S &= \frac{\lambda}{1-\lambda} Z_t + \tau_w W_t^S N_t^S \\
T_t^M &= \tau_p Y_t \\
\tau_w &= 1/(\psi_w - 1) \\
\tau_p &= 1/(\psi_p - 1) \\
Z_t &= z_t Y \\
z_t &= \rho_z z_{t-1} + e_t^z
\end{aligned}
$$

Monetary Policy:

$$
\begin{aligned}
R_t^N &= (1/\beta)\exp(-m_t)(\Pi_t^p)^{\phi} \\
m_t &= \rho_m m_{t-1} + e_t^m
\end{aligned}
$$

Equilibrium conditions

Equilibrium requires market clearing in goods and bond markets, together with consistency between aggregate variables and type-specific allocations. Bond-market clearing requires zero net bond supply, so aggregate bond holdings satisfy $B_t^N = 0$. Aggregate consumption C_t is the population-weighted average of C_t^H and C_t^S, and final output Y_t is allocated to private consumption and to price and wage adjustment costs.

Final goods market:

$$
\begin{aligned}
Y_t = C_t &+ \frac{\eta_p}{2}\left(\frac{P_t}{P_{t-1}} - 1\right)^2 Y_t \\
&+ \lambda\frac{\eta_w}{2}\left(\frac{W_t^{N,H}}{W_{t-1}^{N,H}} - 1\right)^2 Y_t + (1-\lambda)\frac{\eta_w}{2}\left(\frac{W_t^{N,S}}{W_{t-1}^{N,S}} - 1\right)^2 Y_t
\end{aligned}
$$

Bond market: $B_t^N = 0$.

Aggregation condition:

$$
C_t = \lambda C_t^H + (1-\lambda)C_t^S
$$

Reduced nonlinear equilibrium system

Before summarizing the model, I collect the key real variables and inflation rates used below. In particular, $W_t \equiv W_t^N/P_t$ denotes the aggregate real wage, $W_t^H \equiv W_t^{N,H}/P_t$ and $W_t^S \equiv W_t^{N,S}/P_t$ denote type-specific real wages, $\Pi_t^p \equiv P_t/P_{t-1}$ denotes gross price inflation, and Π_t^H, Π_t^S, and Π_t^w denote the corresponding gross wage inflation rates for Type H, Type S, and the aggregate wage index. The aggregate wage markup is defined as $MU_t^w \equiv W_t/MRS(C_t, N_t)$.

The following block collects a reduced set of nonlinear equilibrium conditions that is sufficient for the log-linearization below.

I report only the nonlinear conditions needed to obtain the first-order system below. Conditions omitted here are either redundant or recoverable by symmetry and aggregation.

The nonlinear equilibrium conditions can be summarized as follows:

- Labor Packer:

$$
\begin{aligned}
W_t &= \left[\lambda (W_t^H)^{1-\psi_w} + (1-\lambda)(W_t^S)^{1-\psi_w}\right]^{\frac{1}{1-\psi_w}} \\
N_t &= \left[\lambda (N_t^H)^{\frac{\psi_w - 1}{\psi_w}} + (1-\lambda)(N_t^S)^{\frac{\psi_w - 1}{\psi_w}}\right]^{\frac{\psi_w}{\psi_w - 1}}
\end{aligned}
$$

- Type H:

$$
\begin{aligned}
N_t^H =& N_t \left(\frac{W_t^H}{W_t}\right)^{-\psi_w} \\
C_t^H =& W_t^H N_t^H + Z_t - \frac{\eta_w}{2}\left(\Pi_t^H - 1\right)^2 Y_t \\
\eta_w \Pi_t^H \left(\Pi_t^H - 1\right) =& \psi_w \left\{\frac{1}{MU_t^H} - 1\right\} \frac{W_t^H N_t^H}{Y_t} \\
& + \eta_w \beta \mathbb{E}_t \left[\frac{U_C(C_{t+1}^H, N_{t+1}^H)}{U_C(C_t^H, N_t^H)} \frac{Y_{t+1}}{Y_t} \Pi_{t+1}^H \left(\Pi_{t+1}^H - 1\right)\right] \\
\Pi_t^H =& \frac{W_t^H}{W_{t-1}^H} \Pi_t^p \\
MU_t^H =& \frac{W_t^H}{MRS(C_t^H, N_t^H)}
\end{aligned}
$$

- Type S:

$$
\begin{aligned}
N_t^S =& N_t \left(\frac{W_t^S}{W_t}\right)^{-\psi_w} \\
1 =& \beta \mathbb{E}_t \left[\frac{R_t^N}{\Pi_{t+1}^p} \frac{U_C(C_{t+1}^S, N_{t+1}^S)}{U_C(C_t^S, N_t^S)}\right] \\
\eta_w \Pi_t^S \left(\Pi_t^S - 1\right) =& \psi_w \left\{\frac{1}{MU_t^S} - 1\right\} \frac{W_t^S N_t^S}{Y_t} \\
& + \eta_w \beta \mathbb{E}_t \left[\frac{U_C(C_{t+1}^S, N_{t+1}^S)}{U_C(C_t^S, N_t^S)} \frac{Y_{t+1}}{Y_t} \Pi_{t+1}^S \left(\Pi_{t+1}^S - 1\right)\right] \\
\Pi_t^S =& \frac{W_t^S}{W_{t-1}^S} \Pi_t^p \\
MU_t^S =& \frac{W_t^S}{MRS(C_t^S, N_t^S)}
\end{aligned}
$$

- Firms:

$$
\begin{aligned}
A_t &= \exp(a_t) \\
Y_t &= A_t F(N_t) \\
a_t &= \rho_a a_{t-1} + e_t^a \\
\eta_p \Pi_t^p \left(\Pi_t^p - 1\right) &= \psi_p \left\{ \frac{1}{MU_t^p} - 1 \right\} \\
&\quad + \eta_p \beta \mathbb{E}_t \left[\frac{U_C(C_{t+1}^S, N_{t+1}^S)}{U_C(C_t^S, N_t^S)} \frac{Y_{t+1}}{Y_t} \Pi_{t+1}^p \left(\Pi_{t+1}^p - 1\right) \right] \\
MU_t^p &= \frac{A_t F_N(N_t)}{W_t} \\
D_t &= Y_t - \frac{\eta_p}{2} \left(\Pi_t^p - 1\right)^2 Y_t - W_t N_t
\end{aligned}
$$

- Government:

$$
\begin{aligned}
R_t^N &= \frac{\exp(-m_t)}{\beta} (\Pi_t^p)^\phi \\
m_t &= \rho_m m_{t-1} + e_t^m \\
Z_t &= z_t Y \\
z_t &= \rho_z z_{t-1} + e_t^z
\end{aligned}
$$

- Market clearing and others:

$$
\begin{aligned}
Y_t =& C_t + \frac{\eta_p}{2} (\Pi_t^p - 1)^2 Y_t \\
&+ \lambda \frac{\eta_w}{2} (\Pi_t^H - 1)^2 Y_t + (1-\lambda) \frac{\eta_w}{2} (\Pi_t^S - 1)^2 Y_t \\
C_t =& \lambda C_t^H + (1-\lambda) C_t^S \\
R_t =& \mathbb{E}_t \left[\frac{R_t^N}{\Pi_{t+1}^p} \right] \\
\Pi_t^w =& \frac{W_t}{W_{t-1}} \Pi_t^p \\
MU_t^w =& \frac{W_t}{MRS(C_t, N_t)}
\end{aligned}
$$

Functional specialization and log-linear approximation

From this section onward, I specialize the primitives to the separable CRRA-Frisch utility function and the linear production function:

$$
\begin{aligned}
U(C, N) &= \frac{C^{1-\gamma}}{1-\gamma} - \frac{N^{1+\varphi}}{1+\varphi}, \\
MRS(C, N) &= C^{\gamma} N^{\varphi}, \\
F(N) &= N.
\end{aligned}
$$

In the log-linearized system, lower-case letters denote log deviations from the symmetric zero-inflation steady state; for example, $c_t \equiv \log(C_t/C)$, $n_t \equiv \log(N_t/N)$, and $w_t \equiv \log(W_t/W)$. Inflation variables are log deviations of gross inflation rates, and profits are scaled by steady-state output, $d_t \equiv D_t/Y$. The ex ante real interest rate is $r_t \equiv r_t^N - \mathbb{E}_t \pi_{t+1}^p$, where r_t^N is the log deviation of the gross nominal rate R_t^N. The variables a_t, m_t, and z_t are exogenous deviation processes. In what follows, upper-case MU denotes level markups, whereas lower-case μ denotes their log-linear deviations. Around the symmetric zero-inflation steady state, the quadratic price- and wage-adjustment costs are second-order terms and therefore drop out from the first-order resource constraint, which implies $y_t = c_t$ at first order.

The steady-state values are as follows:

$$
\begin{aligned}
N =& N^S = N^H = 1 \\
Y = C =& C^S = C^H = 1 \\
MU^w =& MU^S = MU^H = 1 \\
\Pi^w =& \Pi^S = \Pi^H = \Pi^p = 1 \\
W =& W^S = W^H = 1 \\
D =& 0
\end{aligned}
$$

The log-linear approximation, with $d_t := D_t/Y$, is then given by:

- Labor Packer:
$$
\begin{aligned}
w_t &= \lambda w_t^H + (1-\lambda) w_t^S \\
n_t &= \lambda n_t^H + (1-\lambda) n_t^S
\end{aligned}
$$

- Type H:
$$
\begin{aligned}
n_t^H &= n_t - \psi_w (w_t^H - w_t) \\
c_t^H &= w_t^H + n_t^H + z_t \\
\pi_t^H &= -(\psi_w/\eta_w)\mu_t^H + \beta \mathbb{E}_t \pi_{t+1}^H \\
\pi_t^H &= w_t^H - w_{t-1}^H + \pi_t^p \\
\mu_t^H &= w_t^H - (\gamma c_t^H + \varphi n_t^H)
\end{aligned}
$$

- Type S:

$$
\begin{aligned}
n_t^S &= n_t - \psi_w(w_t^S - w_t) \\
r_t^N - \mathbb{E}_t\pi_{t+1}^p &= \gamma\mathbb{E}_t\Delta c_{t+1}^S \\
\pi_t^S &= -(\psi_w/\eta_w)\mu_t^S + \beta\mathbb{E}_t\pi_{t+1}^S \\
\pi_t^S &= w_t^S - w_{t-1}^S + \pi_t^p \\
\mu_t^S &= w_t^S - (\gamma c_t^S + \varphi n_t^S)
\end{aligned}
$$

- Firms:

$$
\begin{aligned}
y_t &= a_t + n_t \\
a_t &= \rho_a a_{t-1} + e_t^a \\
\pi_t^p &= -(\psi_p/\eta_p)\mu_t^p + \beta\mathbb{E}_t\pi_{t+1}^p \\
\mu_t^p &= a_t - w_t \\
d_t &= a_t - w_t
\end{aligned}
$$

- Government:

$$
\begin{aligned}
r_t^N &= \phi\pi_t^p - m_t \\
m_t &= \rho_m m_{t-1} + e_t^m \\
z_t &= \rho_z z_{t-1} + e_t^z
\end{aligned}
$$

- Market clearing and others:

$$
\begin{aligned}
y_t &= c_t \\
r_t &= r_t^N - \mathbb{E}_t\pi_{t+1}^p \\
c_t &= \lambda c_t^H + (1-\lambda)c_t^S \\
\pi_t^w &= \lambda\pi_t^H + (1-\lambda)\pi_t^S \\
\pi_t^w &= w_t - w_{t-1} + \pi_t^p \\
\pi_t^w &= -(\psi_w/\eta_w)\mu_t^w + \beta\mathbb{E}_t\pi_{t+1}^w \\
\mu_t^w &= w_t - (\gamma c_t + \varphi n_t)
\end{aligned}
$$

After eliminating redundant equations and the nominal rate r_t^N, Table 1 reports the nonredundant 20-equation representation used in the main text. The system is written in nonredundant form. Some type-specific relations are not reported separately because they are recoverable from symmetry, aggregation, and the displayed household block.

Table 1: Nonredundant 20-equation representation

No.	Name	Equation
1	Type S Euler equation	$r_t = \gamma\mathbb{E}_t\Delta c_{t+1}^S$
2	Type H budget constraint	$c_t^H = w_t^H + n_t^H + z_t$

No.	Name	Equation
3	Aggregate consumption aggregation identity	$c_t = \lambda c_t^H + (1-\lambda)c_t^S$
4	Aggregate labor aggregation identity	$n_t = \lambda n_t^H + (1-\lambda)n_t^S$
5	Aggregate wage Phillips curve	$\pi_t^w = -(\psi_w/\eta_w)\mu_t^w + \beta\mathbb{E}_t\pi_{t+1}^w$
6	Aggregate wage-markup definition	$\mu_t^w = w_t - \gamma c_t - \varphi n_t$
7	Aggregate wage-inflation identity	$\pi_t^w = w_t - w_{t-1} + \pi_t^p$
8	Type H wage Phillips curve	$\pi_t^H = -(\psi_w/\eta_w)\mu_t^H + \beta\mathbb{E}_t\pi_{t+1}^H$
9	Type H wage-markup definition	$\mu_t^H = w_t^H - \gamma c_t^H - \varphi n_t^H$
10	Type H wage-inflation identity	$\pi_t^H = w_t^H - w_{t-1}^H + \pi_t^p$
11	Type H labor demand	$n_t^H = n_t - \psi_w(w_t^H - w_t)$
12	Aggregate wage aggregation identity	$w_t = \lambda w_t^H + (1-\lambda)w_t^S$
13	Aggregate wage-markup aggregation identity	$\mu_t^w = \lambda\mu_t^H + (1-\lambda)\mu_t^S$
14	Aggregate wage-inflation aggregation identity	$\pi_t^w = \lambda\pi_t^H + (1-\lambda)\pi_t^S$
15	Price Phillips curve	$\pi_t^p = -(\psi_p/\eta_p)\mu_t^p + \beta\mathbb{E}_t\pi_{t+1}^p$
16	Price-markup definition	$\mu_t^p = a_t - w_t$
17	Dividend share	$d_t = a_t - w_t$
18	Resource constraint	$y_t = c_t$
19	Production function	$y_t = a_t + n_t$
20	Taylor rule	$r_t = \phi\pi_t^p - \mathbb{E}_t\pi_{t+1}^p - m_t$

For convenience, Table 2 lists the endogenous variables appearing in Table 1, and Table 3 reports the core structural parameters used below.

Table 2: Endogenous variables in the nonredundant system

No.	Meaning	Notation
1	Output	y_t
2	Aggregate consumption	c_t
3	Aggregate labor	n_t
4	Aggregate real wage	w_t
5	Dividend share	d_t
6	Ex ante real interest rate	r_t
7	Price markup	μ_t^p
8	Price inflation	π_t^p
9	Aggregate wage markup	μ_t^w
10	Aggregate wage inflation	π_t^w
11	Type H consumption	c_t^H
12	Type S consumption	c_t^S
13	Type H labor	n_t^H
14	Type S labor	n_t^S
15	Type H real wage	w_t^H

No.	Meaning	Notation
16	Type S real wage	w_t^S
17	Type H wage markup	μ_t^H
18	Type S wage markup	μ_t^S
19	Type H wage inflation	π_t^H
20	Type S wage inflation	π_t^S

Table 3: Core structural parameters

No.	Meaning	Parameter
1	Inverse of elasticity of intertemporal substitution	$\gamma > 0$
2	Inverse of Frisch elasticity	$\varphi > 0$
3	Discount factor	$0 < \beta < 1$
4	Elasticity of substitution across intermediate-goods varieties	$\psi_p \geq 1$
5	Price adjustment cost	$\eta_p \geq 0$
6	Elasticity of substitution across labor varieties	$\psi_w \geq 1$
7	Wage adjustment cost	$\eta_w \geq 0$
8	Taylor-rule coefficient on price inflation	$\phi > 1$
9	Share of Type H households	$0 \leq \lambda < 1$